Futures Margin Requirements

Initial Margin, Daily Settlement, and the Clearing Waterfall

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 2

1 Why Margin Exists

A futures contract has zero value at initiation and full-notional exposure: by replication it is a fully financed spot position, $F_t^T = S_t e^{(r+u-y)(T-t)}$. The **clearinghouse** extends that credit — it novates every trade, becoming buyer to every seller and seller to every buyer — and protects itself by collateralising each position daily. Margin is a performance bond sized by risk, not a down payment sized by price.

2 The Mechanics

2.1 Initial margin

Posted when a position is opened, in cash or Treasuries: enough to cover the loss over a short liquidation horizon at ∼99% confidence. Set by the exchange from recent volatility, revised without notice.

$$\text{IM} \;\approx\; k\,\sigma_{1\text{d}}\sqrt{h} \times \text{notional}, \qquad k \approx 2.5\text{–}3, \;\; h = 1\text{–}2 \text{ days},$$

$\sigma_{1\text{d}}$ daily volatility, h the liquidation horizon, k the coverage multiple. Typically 3–12% of notional; leverage $\approx 1/(k\sigma\sqrt{h})$ falls automatically as volatility rises.

2.2 Daily settlement (variation margin)

Each day the exchange publishes an official **settlement price** F_t and every open position is marked to it:

$$\text{VM}_t \;=\; q \times m \times \big(F_t - F_{t-1}\big),$$

q contracts, m the multiplier. This amount is moved in cash from loser to winner before the next open, re-striking the contract at F_t with zero value. Gains are withdrawable; losses are due now. Over the life of a long opened at F_{t_0}, $\sum_t \text{VM}_t = q\,m\,(F_T - F_{t_0})$ — the entire payoff.

2.3 Maintenance margin and margin calls

When the account balance falls below **maintenance margin** (MM, conventionally 90% of IM), the clearing member issues a **margin call**: restore the balance to the *initial* level, in cash, by the next morning, or be liquidated. Trigger at MM, top-up to IM.

2.4 The clearing chain



IM and VM flow both ways; gross per customer, net per member

2.5 Portfolio margining

Margin is charged on the **net risk of the portfolio**. CME SPAN revalues the book under a grid of price and volatility scenarios and takes the worst case, crediting offsets between correlated positions: a calendar spread is charged 10–20% of the outright; recognised cross-product spreads (crack, crush, NOB) get a partial credit. Capital-efficient — and sized for the offset, not for two independent legs.

3 A Worked Example

Long 10 CL (1,000 bbl each) at \$75.00; IM \$6,000/contract, MM \$5,400/contract. Account funded with exactly IM = \$60,000.

Day	Settle	VM	Balance	vs. MM	Action
0	75.00	—	60,000	+6,000	open, post IM
1	74.20	−8,000	52,000	−2,000	**call**: deposit 8,000 → 60,000
2	74.50	+3,000	63,000	+9,000	excess 3,000 withdrawable
3	73.10	−14,000	49,000	−5,000	**call**: deposit 11,000 → 60,000
4	73.60	+5,000	65,000	+11,000	close: return 65,000

Cash in $60{,}000+8{,}000+11{,}000 = 79{,}000$; cash out 65,000; net loss $14{,}000 = 10\times 1{,}000\times(73.60-75.00)$. Carrying a \$14,000 loss required \$79,000 of liquidity, and on day 3 a 2.5% move put the position one missed wire from liquidation at the low.

The invariant. Margining converts counterparty risk into liquidity risk. Terminal P&L is path-independent; the cash needed to hold the position is not. The binding constraint on a leveraged futures book is the worst drawdown *before* expiry, not the value *at* expiry.